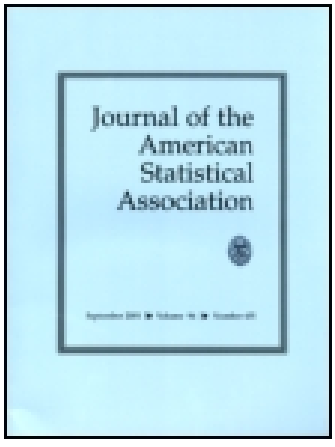
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A Nonparametric "Trim and Fill" Method of Accounting for Publication Bias in Meta-Analysis

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A Nonparametric "Trim and Fill" Method of Accounting for Publication Bias in Meta-Analysis

Sue DUVAL and Richard TWEEDIE

Meta-analysis collects and synthesizes results from individual studies to estimate an overall effect size. If published studies are chosen, say through a literature review, then an inherent selection bias may arise, because, for example, studies may tend to be published more readily if they are statistically significant, or deemed to be more "interesting" in terms of the impact of their outcomes. We develop a simple rank-based data augmentation technique, formalizing the use of funnel plots, to estimate and adjust for the numbers and outcomes of missing studies. Several nonparametric estimators are proposed for the number of missing studies, and their properties are developed analytically and through simulations. We apply the method to simulated and epidemiological datasets and show that it is both effective and consistent with other criteria in the literature.

KEY WORDS: Data augmentation; File drawer problem; Funnel plot; Lung cancer; Meta-analysis; Missing studies; Passive smoking; Publication bias.

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1. INTRODUCTION

1.1 The Publication Bias Problem

There has been an enormous recent increase in the use of meta-analysis as a statistical technique for combining the results of many individual analyses in areas of clinical trials, epidemiology, sociology, and psychology (Cooper and Hedges 1994; Hedges and Olkin 1985; Olkin 1992). In all of these contexts, one well-documented concern is the need to collect all relevant studies, both published and unpublished, if the subsequent inferences are to be valid (Begg 1994; Begg and Mazumdar 1994; Dear and Begg 1992; Egger, Smith, Schneider, and Minder 1997; Gleser and Olkin 1996; Hedges 1992; Iyengar and Greenhouse 1988).

The use of a nonrepresentative proportion of significant studies, or studies differentially giving results in a "positive" direction, will lead to a nonrepresentative set of studies in the meta-analysis dataset. A standard meta-analysis model will then result in a conclusion biased twoard significance or positivity. This is particularly problematic for a meta-analysis whose data come solely from the published scientific literature, but studies may be suppressed for many other reasons than failure to be published (Cooper 1998; Givens, Smith, and Tweedie 1997) and the discussion thereto, Misakian and Bero 1998). This phenomenon has become known as the "publication bias," or the "file-drawer problem" (Iyengar and Greenhouse 1988).

In this article we describe a new way to account for the magnitude of this problem. Our approach has two steps. First, we estimate the number of missing studies using methods based on symmetry assumptions. These are simple to implement, in contrast to most existing methods, and they appear in practice to pick up the "missing studies" indicated visually by funnel plots (Light and Pillemer 1984), as described in Section 1.2. Second, we impute the missing values using a "trim and fill" approach that enables us to derive estimates of the overall effect on the inferences in the meta-analysis due to the publication bias.

We show that this dual approach works well in simulation studies and illustrate its use on an epidemiological example. Rather than merely testing for the possibility of publication bias (as in, say, Begg 1994 or Egger et al. 1997), the trim and fill algorithm appears to give a method of evaluating and also adjusting for the possibility of publication bias, so that one can tell whether or not the magnitude of the problem warrants consideration.

1.2 Funnel Plots and Publication Bias

Several methods have been proposed to detect the existence of publication bias in a meta-analysis. Perhaps the most common is the funnel plot (Light and Pillemer 1984) and related graphical methods for visually determining the existence of missing studies (Galbraith 1988).

Figure 1(a) shows a funnel plot using data from a simulation of 35 studies, each of which supplies an estimate Y_i of the effect in question in the ith study, and an estimate of the variance σ_i^2 within that study. The funnel plots that we use depict σ_i^{-1} graphed against Y_i, so that the most precise estimates (e.g., those from the largest studies) are at the top of the funnel, and those from less precise or smaller studies are at the base of the funnel. The data were generated as in Section 4 with $Y_i \sim \mathrm{N}(0, \sigma_i^2)$, so that the true mean Δ here is 0; the sample used has meta-analyzed estimate (using the random-effects model described later) $\hat{\Delta} = .080$ with a 95% confidence interval of $(-.018, .178)$, which is not significant at the usual 5% level.

The fact that there is a "funnel" shape is not based on any detailed modeling; it relies on two empirical observations:

(F1) The variances σ_i^2 of studies in a meta-analysis are not identical, but are distributed in such a way that there are fewer precise studies and rather more imprecise studies. This appears to be borne out by data in, for example, Light and Pillemer (1984, chap. 3) among many others.

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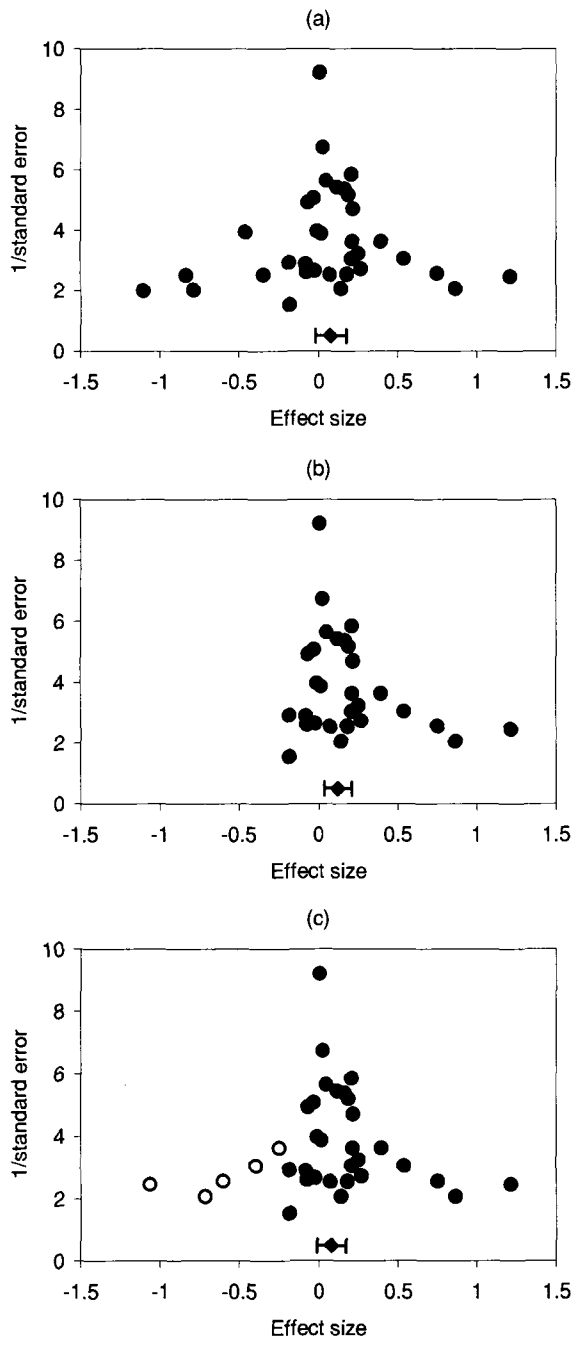


Figure 1. Funnel plots of (a) 35 simulated studies, with an overall effect size of .080 with a 95% confidence interval of (−.018, .178); (b) with the five "left-most" studies suppressed with an overall effect size estimated at .124 with a 95% confidence interval of (.037, .210); and (c) "Filled" funnel plot with overall effect size estimated using trim and fill as .082, with a 95% confidence interval of (−.011, .176).

(F2) At any fixed level of σ_i^2, studies are symmetrically distributed around the true mean Δ. This is justified if we use the normal model in Section 2.1, but clearly is a much weaker assumption than such normality.

To illustrate the effect of publication bias, in Figure 1(b) we show the same funnel plot as in Figure 1(a), but with the "left-most" five studies suppressed. This gives the typical pattern that is taken to indicate publication bias (Light and Pillemer 1984). The assumption is that, whether because of editorial policy or author inaction or other reasons, these articles (which show, say, no significance, or perhaps the reverse effect, namely $\Delta < 0$, from that envisaged when carrying out the studies, namely $\Delta > 0$) are the ones that might not be published (see Cooper 1998, pp. 54–55). Such suppression will affect the estimate of Δ. Indeed, in this example it increases $\hat{\Delta}$ to .124 with a 95% confidence interval of (.037, .210), which is now significant at the 5% level. In other words, suppressing these five studies leads to an incorrect inference by usual standards, and increases the estimated mean by almost 60%.

The evaluation of funnel plots is subjective in much of the literature. Other, more quantitative methods of detecting publication bias have been proposed, ranging from a simplistic method-of-moments approach (Sugita 1992, 1994) to more sophisticated methods such as a rank correlation test (Begg 1994; Begg and Mazumdar 1994) based on the form of a biased funnel plot as in Figure 1(b), a method based on p values (Gleser and Olkin 1996), a regression-based test (Egger et al. 1997) based on the Galbraith (1988) version of a funnel plot, and a Bayesian approach assuming a known selection parameter (Eberly and Casella 1999).

These tests are all designed to detect rather than adjust for publication bias. There also exist several quantitative methods that estimate the number of missing studies and, by explicitly modeling the probability of publication, provide estimates of the effect of the missing studies on the overall effect size Δ (Dear and Begg 1992; Givens et al. 1997; Hedges 1992; Smith, Givens, and Tweedie 1999). All are complex and highly computer-intensive to run, and Dear and Dobson (1997) noted that "previous methods have not been much used ... and ... the value of any new statistical methodology depends, in part, on the extent to which it is adopted." They also noted that "the culture of meta-analysis has traditionally favoured very simple methods." We now develop a simple technique that seems to meet many of the objections to other methods.

2. ADJUSTING FOR PUBLICATION BIAS

2.1 Fixed- and Random-Effects Models

We first describe a standard structure for a meta-analysis in the absence of publication bias. We assume that we have n individual studies, all of which are addressing the same problem, and that there is some global "effect size" Δ (such as log relative risks, risk differences, or log mortality ratios in clinical or epidemiological trials, or differences in performance in sociological experiments) that is relevant to the overall problem and that each study attempts to measure.

For each $j = 1, \ldots, n$, study j produces an *effect size* Y_j that estimates Δ and an *estimated within-study variance* σ_j^2. The random-effects (RE) model commonly used to combine the Y_j is then

$$Y_j = \Delta + \beta_j + \varepsilon_j, \tag{1}$$

where $\beta_j \sim \mathrm{N}(0, \tau^2)$ is introduced to account for heterogeneity between studies and $\varepsilon_j \sim \mathrm{N}(0, \sigma_j^2)$ represents the within-study variability of study j. The RE approach has

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been argued (NRC Report 1992) to be preferable to the fixed-effects (FE) model, which assumes that $\tau^2 = 0$, so that any heterogeneity between studies is purely random.

This model leads, through normal theory, to the meta-analyzed estimate of Δ given by $\hat{\Delta} = \sum Y_j w_j / \sum w_j$, where $w_j = (\sigma_j^2 + \tau^2)^{-1}$ and with $\text{var}[\hat{\Delta}] = 1/\sum w_j$. In fitting this estimator, it is usually assumed that the σ_j^2 are known. In the FE model we take $\tau^2 = 0$; in the RE model there are various moment-based and maximum likelihood approaches giving estimates of τ^2 (Biggerstaff and Tweedie 1997), the most common being the DerSimonian–Laird estimator (DerSimonian and Laird 1986), which we use in Section 5 and Section 6. This is given by

$$\hat{\tau}^2_{\text{DL}} = \max\left\{0, [Q - (K-1)] \Big/ \left[S_1 - \left(\frac{S_2}{S_1}\right)\right]\right\}, \quad (2)$$

where $Q = \sum \sigma_j^{-2}(Y_j - \hat{\Delta}_{\text{FE}})^2$; here $\hat{\Delta}_{\text{FE}} = (\sum \sigma_j^{-2} Y_j)/\sum \sigma_j^{-2}$ is the weighted least squares estimator of Δ using the FE method, and $S_r = \sum \sigma_j^{-2r}$.

2.2 Modeling Publication Bias

Now we modify the foregoing model to account for publication bias. We assume that in addition to the n observed studies, there are an additional k_0 relevant studies that are not observed, due to publication bias. The value of k_0, and the effect sizes that might have been found from these k_0 studies, are unknown and must be estimated; uncertainty about these estimates must be reflected in the final meta-analysis inference.

The frequentist models of Dear and Begg (1992) and Hedges (1992) and the Bayesian models of Givens et al. (1997) and Smith et al. (1999) estimate both k_0 and the positions of the missing studies using the aforementioned normal models together with a model of the probabilities that studies are missing in different p-value intervals. In contrast, our approach is nonparametric and relies only on symmetry assumptions, which are certainly satisfied by both the FE and RE models.

To describe this, we first recall the standard assumption behind the Wilcoxon test for a collection $X_i, i = 1, \ldots, N$ of random variables (Marascuilo and McSweeney 1977). If the hypothesis is that the median of each of the $X_i = 0$, then each observed value is assumed to have arisen by first generating $|X_i|$ and then generating the sign of X_i according to an independent set of Bernoulli variables taking values $-1, 1$ with equal probability. If r_i denotes the rank of $|X_i|$, and if

$$W_N^+ = \sum_{X_i > 0} r_i$$

is the sum of the ranks associated with the positive X_i, then we say that W_N^+ has a Wilcoxon distribution.

Now let us suppose that there are originally $n + k_0$ studies with this structure and that k_0 values from the set X_j have been suppressed, leaving a set of n observed studies. Our key assumption is that *the suppression has taken place in such a way that the k_0 values of the X_j with the most extreme negative ranks have been suppressed.* This might be expected to lead to a truncated funnel plot such as we have in Figure 1(b), for example. We call this model for the overall set of studies the *suppressed Bernoulli model.*

We are now left with n observed values, and these remaining $|X_i|$ are reranked as r_i^*: these ranks run now from 1 to n. We consider three statistics based on this set of observed ranks. First, we let $\gamma^* \geq 0$ denote the length of the rightmost run of ranks associated with positive values of the observed X_i; that is, if h is the index of the most negative of the X_i, and if r_h^* is its absolute rank, then $\gamma^* = n - r_h^*$. Second, we consider the Wilcoxon statistic for the observed set of variables omitting the suppressed variables. To distinguish the suppressed dataset from a set of random variables, each of which is symmetric, we define the "trimmed" rank test statistic for the observed n values as

$$T_n = \sum_{X_i > 0} r_i^*. \quad (3)$$

The distributions of γ^* and T_n depend on k_0, although we omit this in the notation.

Based on these quantities, we define three estimators of k_0, given by

$$R_0 = \gamma^* - 1, \quad (4)$$

$$L_0 = \frac{4T_n - n(n+1)}{2n - 1}, \quad (5)$$

and

$$Q_0 = n - \frac{1}{2} - \sqrt{2n^2 - 4T_n + \frac{1}{4}}. \quad (6)$$

The forms of L_0 and Q_0 are based on method-of-moments considerations developed in Section 3.2. The main analytic results of this article are as follows, which are proved in the Appendix.

Theorem 1. Under the assumption that the median of the original X_i is 0, and under the suppressed Bernoulli model, the following hold:

(a) The estimator R_0 has mean and variance given by

$$E[R_0] = k_0, \qquad \text{var}[R_0] = 2k_0 + 2. \quad (7)$$

(b) The estimator L_0 has mean and variance given by

$$\begin{aligned} E[L_0] &= k_0 - k_0^2/(2n-1), \\ \text{var}[L_0] &= 16\ \text{var}(T_n)/(2n-1)^2, \end{aligned} \quad (8)$$

where

$$\begin{aligned} \text{var}(T_n) = 24^{-1}[&n(n+1)(2n+1) + 10k_0^3 + 27k_0^2 \\ &+ 17k_0 - 18nk_0^2 - 18nk_0 + 6n^2k_0]. \end{aligned} \quad (9)$$

(c) The estimator Q_0 has mean and variance given (approximately) by

$$\begin{aligned} E[Q_0] &\approx k_0 + \frac{2\ \text{var}[T_n]}{[(n-\frac{1}{2})^2 - k_0(2n - k_0 - 1)]^{3/2}}, \\ \text{var}[Q_0] &\approx \frac{4\ \text{var}[T_n]}{(n-\frac{1}{2})^2 - k_0(2n - k_0 - 1)}. \end{aligned} \quad (10)$$

Remark. If we consider the asymptotic situation where n is large and k_0 is of smaller order than n, then

$$E[L_0] \sim k_0, \qquad E[Q_0] \sim k_0 + \frac{1}{6}, \qquad \text{and } E[R_0] = k_0$$

and

$$\text{var}[L_0] \sim n/3, \qquad \text{var}[Q_0] \sim n/3, \qquad \text{and var}[R_0] = o(n).$$

Hence both L_0 and Q_0 will have similar behavior, although in practice Q_0 usually seems to be of greater magnitude than L_0 and occasionally appears to give excessively large values. Although var$[L_0]$ is asymptotically larger than var$[R_0]$, comparing the analytic forms of their mean squared errors (MSEs) based on (7), (8), and (10), we find that for any fixed n, the MSE of L_0 is actually smaller than that of R_0 for larger values of k_0. The region in which L_0 is better in MSE than R_0 is very closely approximated by

$$k_0 \geq n/4 - 2.$$

Thus if the number of missing studies is estimated to be greater than about 25% of those observed, then we suggest that using L_0 might be preferable.

Remark. Because we have an expression for the bias of L_0, we can in principle remove this to at least $O(n^{-2})$. If we define

$$L_1 = L_0[1 - (1/2n)] + L_0^2/(2n-1) - \frac{1}{6} - 1/4n, \quad (11)$$

then it is possible to check that, again for n large and k_0 fixed and smaller than n,

$$E[L_1] = k_0 + O(n^{-2}).$$

Using L_1 seems to make only marginal difference in simulations (Duval 1999).

3. ESTIMATORS OF k_0

We now assume that the median of the X_i is known, and take it as 0. This restriction is removed later.

3.1 The "Rightmost Run" Estimator R_0

We first examine the simple estimator R_0 defined by (4). We begin by considering the sequence of signed ranks in the original (unsuppressed) dataset. As a specific example, suppose that the original set of studies is of size $n + k_0 = 14$, with the ranks and their associated signs being

$$-1, -2, 3, 4, 5, -6, -7, 8, 9, -10, 11, 12, -13, 14.$$

Now suppose that we suppress $k_0 = 3$ studies, which (because we suppress the three most negative studies) means that we suppress those corresponding to $-13, -10, -7$. The signed ranks after ranking the remaining 11 studies are

$$-1, -2, 3, 4, 5, -6, 7, 8, 9, 10, 11.$$

We can consider this as comprising two groups: the set {7, 8, 9, 10, 11}, which were "promoted" to be the top five ranks, with positive sign, due to the suppression, and the remaining six ranks, which still have the same random permutation as before. Thus in this case the run of positive ranks is of length five, and $\gamma^* = 5$. But we can see that γ^* is just the number of positive terms that we need to reach the first negative value that is *not* suppressed, which is the $(k_0 + 1)$st negative term in the original sequence, and hence γ^* is a negative binomial variable corresponding to a sequence of simple Bernoulli coin-tosses.

Using the negative binomial form (Casella and Berger 1990, p. 96), we thus have immediately the following result, which gives (7).

Theorem 2. The distribution of the estimator R_0 is given by

$$P(R_0 = m) = \binom{k_0 + m + 1}{m + 1} \frac{1}{2}^{k_0 + m + 2}. \quad (12)$$

Remark. The major drawback to this estimator is that it is nonrobust to the existence of a relatively isolated negative term at the right hand end of the sequence of ranks. Of course, this is correct given the exact application of the suppression hypothesis that we have used, but in practice we may find funnel plots with one such isolated value and the appearance of a gap in the next run of positive values. The use of R_0 may be rather conservative in this case, and the next set of estimators, based on T_n, seem more robust to this type of departure from the Bernoulli suppression hypothesis.

3.2 The Linear and Quadratic Estimators L_0 and Q_0

We now find the first and second moments of T_n and show that we can use these to derive the forms L_0 and Q_0 as in (5) and (6). The proof of the next result is given in the Appendix.

Theorem 3. For given fixed k_0, under the suppressed Bernoulli model,

$$T_n = n + (n-1) + \cdots + (n - \gamma + 1) + W^+_{n-\gamma}, \quad (13)$$

where W^+_N has the Wilcoxon distribution and γ has the negative binomial distribution

$$P(\gamma = m) = \binom{k_0 + m - 1}{m} \frac{1}{2}^{k_0 + m}. \quad (14)$$

The first two moments of this mixture distribution are

$$E[T_n] = [n(n+1) + k_0(2n - k_0 - 1)]/4 \quad (15)$$

and var$[T_n]$ given by (9).

We first consider the method-of-moments estimator for k_0, which follows by solving (15). This gives

$$Q_0 = \left(n - \frac{1}{2}\right) - \sqrt{\left(n - \frac{1}{2}\right)^2 - 4[T_n - n(n+1)/4]}, \quad (16)$$

which gives (6). In defining this estimator, we assume that only the negative root of the quadratic (15) is relevant. The

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other root leads to a (rounded) estimate of k_0, that is always as large as n. Our choice of Q_0 thus implicitly requires a belief that we are in fact observing at least one-half of the real studies, which seems the more realistic situation.

More important, for Q_0 to be well defined, the discriminant $2n^2 - 4T_n + \frac{1}{4}$ must be nonnegative; that is, we must have $T_n < n^2/2 + \frac{1}{16}$ for this estimator to be defined. Simulations show that when k_0 is large with relation to n, and when n itself is small, this may be violated quite frequently. Thus, although we do evaluate properties of Q_0, it clearly has drawbacks in practice. We note, however, that the discriminant is negative only for large values of T_n, and that at the point where it is 0, our estimate would be $n - \frac{1}{2}$. It is thus logical to use this latter, rather large estimate as a lower bound when Q_0 is complex.

Because of these concerns, we now consider a second estimator based on (15). If we ignore the quadratic term involving k_0 in (15), which is relatively small, at least for large n, then from the linearization

$$T_n - n(n+1)/4 \approx (2n-1)k_0/4,$$

we find the explicit linear estimator

$$L_0 = \frac{4(T_n - n(n+1)/4)}{2n-1}, \tag{17}$$

as given in (5).

Remark. The negative binomial forms here are reminiscent of those used by Eberly and Casella (1999) and Givens et al. (1997). However, the parameters of our negative binomials are explicit in terms of k_0, whereas theirs are related to the probabilities of suppression in different p-value regions and do not seem to be related to our method here.

4. PROPERTIES OF THE ESTIMATORS

In using any of the estimators in Sec. 3, we round to the nearest nonnegative integer, because algorithmically we need to trim whole studies rather than fractions of studies. Thus in practice we use

$$\begin{aligned} R_0^+ &= \max\{0, R_0\}, \\ L_0^+ &= \left[\max\left\{0, L_0 + \frac{1}{2}\right\}\right], \\ Q_0^+ &= \left[\max\left\{0, Q_0 + \frac{1}{2}\right\}\right], \end{aligned} \tag{18}$$

where $[x]$ is the integer part of x.

We now show, using simulations, that the means and variances of these estimators, which we call "empirical" versions, are acceptably close to those in Theorem 1 for the ranges of n, k_0 that we address, and thus the use of the theoretical values seems justified. To carry out the simulations, we generated for each of the following cases, 5,000 sets of $N = n + k_0$ normal variates, each with mean 0 and variance σ_i^2, with the σ_i taken from a $\Gamma(3, \frac{1}{9})$ density.

Figure 2 shows the means of the simulations for R_0^+, L_0^+, and Q_0^+ compared to the theoretical means of R_0, L_0, and Q_0 from Theorem 1. Over the range of N that we have compared, when $k_0 = 5$ or 10, there is very little difference between the analytic and empirical forms in general; the means of L_0 and L_0^+ differ by no more than .6 in the former case and by no more than .16 in the latter case, for example. We see that R_0 is even more stable and that the analytic forms for Q_0 (which, we recall, were only approximate) are also accurate except in the relatively extreme case where $n = 15$ and $k_0 = 10$.

The variances of the empirical estimators are also generally well approximated by the analytic forms in the cases where $k_0 = 5$ or 10. The exception is again when $k_0 = 10$

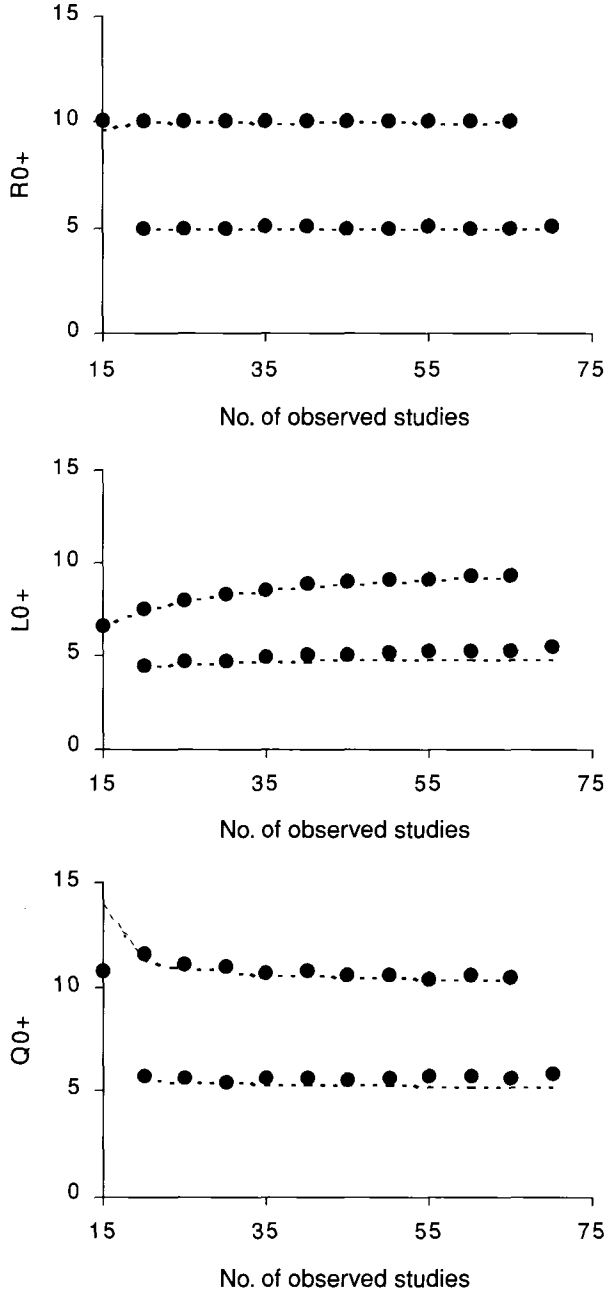


Figure 2. Analytic means (lines) and simulated means (symbols) for each of the estimators shown, at values of k_0 = 5 and 10.

for Q_0; here we seem to need n to be greater than around 35 for the approximations to hold.

When $k_0 = 0$, truncating to ensure nonnegative empirical estimators leads to means that are higher than the analytic forms, as we must expect. In another article (Duval and Tweedie 1999) we develop tests of the hypothesis that $k_0 = 0$ based on the foregoing distributional properties and find that even with the truncations, under $k_0 = 0$ approximately 95% of the values of R_0^+ are less than or equal to 2, and approximately 95% of the values of L_0^+ are less than or equal to 3–5 when $n \leq 50$. These tests were utilized further by Sutton, Duval, Tweedie, Abrams, and Jones (1999).

5. THE ITERATIVE "TRIM AND FILL" ALGORITHM

The foregoing results show the behavior of our estimators when the value of Δ is known. The more difficult situation that arises in meta-analysis is caused because Δ is unknown, so that the number and position of any missing studies is correlated with the true value of Δ. To handle this situation, we develop an iterative algorithm. Using the estimators in Sec. 2.2, we proceed as follows:

Step 1. Estimate $\hat{\Delta}^{(1)}$ using the RE estimator. Construct the initial set of centered values

$$Y_i^{(1)} = Y_i - \hat{\Delta}^{(1)}, \qquad i = 1, \ldots, n,$$

and estimate $\hat{k}_0^{(1)}$ using one of the estimators (say L_0^+) applied to the set $Y_i^{(1)}$.

Step 2. Relying on a symmetry argument, remove $k_0^{(1)}$ values from the right end of the set of values Y_i, and estimate $\hat{\Delta}^{(2)}$ based on the trimmed "symmetric" set of $n - \hat{k}_0^{(1)}$ values; that is, $\{Y_1, \ldots, Y_{n-\hat{k}_0^{(1)}}\}$. Construct the next set of centered values

$$Y_i^{(2)} = Y_i - \hat{\Delta}^{(2)}, \qquad i = 1, \ldots, n,$$

and estimate $\hat{k}_0^{(2)}$ from the set $Y_i^{(2)}, i = 1, \ldots, n$.

Step 3. Remove $\hat{k}_0^{(2)}$ values from the right end; reestimate $\hat{\Delta}^{(3)}$ based on the leftmost set of $n - \hat{k}_0^{(2)}$ values; construct $Y_i^{(3)} = Y_i - \hat{\Delta}^{(3)}, i = 1, \ldots, n$; estimate $\hat{k}_0^{(3)}$ from $Y_i^{(3)}$.

Step 4. Continue until an iteration J, where $\hat{k}_0^{(J)} = \hat{k}_0^{(J-1)} := \hat{k}_0$, at which point we also have $\hat{\Delta}^{(J)} = \hat{\Delta}^{(J-1)}$.

Step 5. As a final step, "fill" the funnel plot by using the imputed symmetric values

$$Y_j^* = 2\hat{\Delta}^{(J)} - Y_{n-j+1}, \qquad j = 1, \ldots, \hat{k}_0$$

with associated imputed standard errors

$$\sigma_j^* = \sigma_{n-j+1}, \qquad j = 1, \ldots, \hat{k}_0.$$

Estimate the final value of Δ using the full augmented dataset by

$$\hat{\Delta}^{(F)} = \frac{\sum_1^n Y_i w_i + \sum_1^{\hat{k}_0} Y_i^* w_i^*}{\sum_1^n w_i + \sum_1^{\hat{k}_0} w_i^*} \tag{19}$$

with imputed 95% confidence interval given by

$$\left[\hat{\Delta}^{(F)} - \frac{1.96}{\sqrt{\sum_1^n w_i + \sum_1^{\hat{k}_0} w_i^*}}, \; \hat{\Delta}^{(F)} + \frac{1.96}{\sqrt{\sum_1^n w_i + \sum_1^{\hat{k}_0} w_i^*}}\right]. \tag{20}$$

Here $w_i = [\sigma_i^2 + \tau_F^2]^{-1}$ and $w_i^* = [(\sigma_i^*)^2 + \tau_F^2]^{-1}$, and τ_F^2 is estimated from the entire filled dataset $\{Y_1, \ldots, Y_n, Y_1^*, \ldots, Y_{\hat{k}_0}^*\}$, using the DerSimonian–Laird estimator.

We make the following observations on this algorithm:

a. Note that finding $\hat{\Delta}^{(J)}$ in step 4 is possible only if $\hat{k}_0 \leq n - 1$, and in practice we truncate the algorithm if this value is reached.
b. When we have a FE model, it is clear that the algorithm must converge, because the $\hat{\Delta}^{(j)}$ must decrease by definition, and so $\hat{k}_0^{(j)}$ increases. In the real (nonsimulated) examples in Duval (1999), we found that convergence never took more than four iterations. However, if the data are fitted by a random-effects model, then it is possible that the $\hat{\Delta}^{(j)}$ may oscillate, and so $\hat{k}_0^{(j)}$ also will oscillate. This appears to be rare in simulations, but the possibility must be considered.
c. The final estimate reflects the variability that should be imputed in the missing studies: thus we are not only adjusting the mean but also the confidence interval for these imputed values. In fact, one can check that in the FE case the sum in (19) depends only on the $n - \hat{k}_0$ values in the trimmed data set, so the final mean in this context does not depend on the location but only on the number of imputed missing values. In the RE case this is not so, since the imputed (filled) values may change the estimate of τ^2.

Properties of the iterated algorithm were explored in more detail in another article (Duval and Tweedie 2000), where we showed, using simulations, that the iterative step does not seem to adversely affect the performance of the method.

6. TWO EXAMPLES

We now apply the method to two different examples the simulated dataset in Section 1.2, where we know the underlying structure, and a meta-analysis of 35 studies of the effect of environmental tobacco smoke (ETS) on lung cancer, studied by Givens et al. (1998) using a relatively complex Bayesian approach. We compare the results obtained with those from three other methods: the estimate of the number of unreported studies calculated by the "simple" method of equation (2) of Gleser and Olkin (1996), which is comparable in calculation effort to our estimates of k_0, and the tests for publication bias of Begg (1994) and Egger et al. (1997).

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Table 1. Effect of Publication Bias Using Estimators of k_0 for the Data in Figure 1

Estimator	Imputed n	Value of k_0 (SE)	No. of iterations	$\hat{\Delta}$	95% confidence interval
Original data	35	0		.080	(−.018, .178)
Suppressed data	30			.124	(.037, .210)
R_0^+	34	4 (3.2)	2	.092	(−.001, .185)
L_0^+	35	5 (3.7)	5	.082	(−.011, .176)
Q_0^+	36	6 (4.6)	6	.079	(−.013, .170)

6.1 Example: Simulated Data

In the simulated data shown in Figure 1, we know that five studies have been suppressed, and that this changes a statistically insignificant result on the original data to a statistically significant result if publication bias is not considered. On this dataset, the method of Gleser and Olkin (1996) estimates that there are 11 missing studies. The two-sided p values from the methods of Begg (1994) and Egger et al. (1997) are .14 and .004, indicating that the null hypothesis of no suppression is not supported. Note that Begg (1994) pointed out that his test is not powerful, and a value such as this should be taken as an indication of the possible existence of publication bias.

Table 1 shows how the various estimators behave on this example. As is often found in examples, we have $R_0^+ \leq L_0^+ \leq Q_0^+$. There is relatively little difference on the filled value of Δ and its confidence interval, no matter which estimator is used to fill. Figure 1(c) is filled using the actual number, which is equivalent to using L_0^+.

6.2 Example: Lung Cancer and Environmental Tobacco Smoke

Givens et al. (1997) used a Bayesian approach to estimate and adjust for publication bias in a set of 35 studies that assesses the risk of lung cancer in nonsmoking women exposed to spousal smoking. The data consist of relative risks and associated confidence intervals, and also were studied by Lee (1992) and Mengersen, Tweedie, and Biggerstaff (1995).

We find that $R_0^+ = L_0^+ = 8$, and $Q_0^+ = 9$. In this case the two-sided p values from the methods of Begg (1994) and Egger et al. (1997) are .16 and .05, also indicative of publication bias. In contrast, the method of Gleser and Olkin (1996) estimates that $k_0 = 0$; this is because their method is very susceptible to just one study with a large p value against the null hypothesis (in this case due to a Y_i that is only slightly negative but has a very small variance).

After filling eight studies, in accord with R_0^+ or L_0^+, the overall relative risk moves from 1.20 [1.08, 1.34] to 1.12 [1.01, 1.25], almost identical to the more complicated procedure of Givens et al. (1997) and the more subjective analysis of Mengersen et al. (1995). This is illustrated in Figure 3, where the original data are shown as solid symbols and the "filled" data as open symbols. Figure 3(a) uses the appropriate effect size in the funnel plot, whereas 3(b) gives the meta-analyzed estimates and confidence intervals in the original scale.

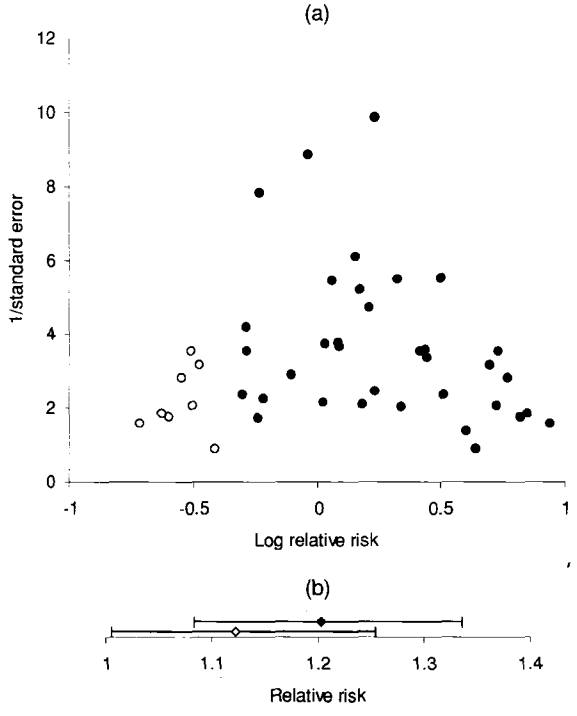


Figure 3. (a) Funnel plot of log relative risks of lung cancer associated with ETS; (b) overall relative risk and 95% confidence interval before and after allowing for publication bias.

7. DISCUSSION

A meta-analysis based on only a subset of all relevant studies may result in biased conclusions. It is a common belief, backed by several empirical assessments, that studies are not uniformly likely to be published in scientific journals (Cooper 1998, pp. 54–55; Dickersin, Min, and Meinert 1992). Easterbrook et al. (1991) suggested that statistical significance is a major determining factor of publication. Some researchers (e.g., students with masters or doctoral theses) may not submit a nonsignificant result for publication, and editors may not publish nonsignificant results even if they are submitted (British Medical Journal 1983).

Evaluating this effect is difficult, because the missing studies influence the mean estimated in the meta-analysis. As DuMouchel and Harris (1997) noted, "attempts to assess publication bias beyond simple graphs like the funnel plot seem to involve a tour de force of modeling, and as such are bound to run up against resistance from those who are not statistical modeling wonks."

The method that we propose here avoids the need for such a cumbersome model by using a simple approach to calculating k_0 (in terms of computational burden and as-

sumptions), and by applying this iteratively as we trim and fill the data. Our method is based on a particular suppression mechanism. Other suppression mechanisms have been suggested in the literature: suppression of all studies with p values above a certain level (e.g., of all insignificant studies), as discussed by Gleser and Olkin (1996), or such suppression according to various probabilities depending on the level of significance (Dear and Begg 1992; Eberly and Casella 1999; Hedges 1992; Givens et al. 1997; Smith et al. 1999). A number of authors (e.g., DuMouchel and Harris 1997) have pointed out that the simple p-value scenario is rather simplistic, because it fails to acknowledge the role of other criteria, such as size of study, in the decision as to whether a study is published. Misakian and Bero (1998) have studied publications in passive smoking (see Sec. 6) and found that p values are probably the critical determinant in delaying publication, but study size does appear to have some effect.

Our mechanism allows for many of these issues. It assumes that (one-sided) insignificance is, in the main, likely to lead to suppression, but if such a poor p value is from a large but only slightly "negative" study, as may well happen, then this is not likely to be the type of p value actually omitted. In this way, the Bernoulli suppression scenario seems appropriate. Of course, in many cases suppression according to p values is similar to our Bernoulli suppression assumption; when the σ_i are constant, the most negative effect sizes have the largest one-sided p values.

Further study is under way to determine how robust the various estimators are to other suppression mechanisms (Duval and Tweedie 2000). Initial simulations show that L_0 is more robust than R_0 against certain data configurations that might occur under other circumstances. For example, if there is just one very negative value of X_i followed by a missing collection of studies (which violates the exact assumption that we have made, but not the spirit of publication bias), then R_0 must be zero, although L_0 may be nonzero.

The results that we have reported indicate that the nonparametric estimators for the number of missing studies work well both for data simulated under the Bernoulli assumption and in real examples, giving realistic estimates from R_0^+ or L_0^+ in particular. Comparisons of these estimators have been given in more detail by Duval (1999), Duval and Tweedie (1999), and Sutton et al. (1999). In the examples examined there, the trim and fill method also matches the subjective impression of bias given by funnel plots and appears to give results consistent with those of Begg (1994) and Egger et al. (1997).

Of course, we do not actually know that the filled studies match reality in any close way. But in the sense that we can see the potential for incorrect conclusions if publication bias does exist, the trim and fill method does seem to provide a robust diagnostic method to enable us to decide which meta-analyses are safe from this bias, and which ones must be treated with considerable caution.

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APPENDIX: PROOFS

Proof of Theorem 1

a. This is proved directly from Theorem 2.

b. For the estimator L_0, we have, using (A.6),

$$E[L_0] = \frac{4}{[2n-1]}\, E[T_n - E[W_n^+]]$$

and

$$\begin{aligned}\operatorname{var}[L_0] &= \left(\frac{4}{2n-1}\right)^2 \operatorname{var}[T_n - E[W_n^+]] \\ &= \left(\frac{4}{2n-1}\right)^2 \operatorname{var}[T_n]. \end{aligned} \tag{A.1}$$

c. For the estimator Q_0, we write $Q_0 = f(X) = (n-\frac{1}{2}) - \sqrt{(n-\frac{1}{2})^2 - 4X}$ where $X = T_n - E[W_n^+]$ and use a Taylor series expansion about $\mu = E[X]$. Using the first-order approximation $E[Q_0] \approx f(\mu)$, we get from (A.6) that

$$\begin{aligned}E[Q_0] &\approx \left(n-\frac{1}{2}\right) - \sqrt{\left(n-\frac{1}{2}\right)^2 - 4(E[T_n] - E[W_n^+])} \\ &= \left(n-\frac{1}{2}\right) - \sqrt{\left(n-\frac{1}{2}\right)^2 - k_0(2n-k_0-1)}. \end{aligned} \tag{A.2}$$

Adding a second-order term, we use the approximation $E[Q_0] \approx f(\mu) + \frac{1}{2}f''(\mu)\operatorname{var}[X]$ to give

$$E[Q_0] \approx k_0 + \frac{2\operatorname{var}[T_n]}{[(n-\frac{1}{2})^2 - k_0(2n-k_0-1)]^{3/2}}. \tag{A.3}$$

To derive the variance of Q_0, we use the approximation that $\operatorname{var}[Q_0] \approx [f'(\mu)]^2\operatorname{var}[X]$, so that

$$\begin{aligned}\operatorname{var}[Q_0] &\approx \left[\frac{2}{\sqrt{(n-\frac{1}{2})^2 - k_0(2n-k_0-1)}}\right]^2 \operatorname{var}[T_n] \\ &= \frac{4\operatorname{var}[T_n]}{(n-\frac{1}{2})^2 - k_0(2n-k_0-1)}. \end{aligned} \tag{A.4}$$

Proof of Theorem 3

By removing the "top" k_0 negative ranks, we are left with matching positive ranks, and we denote the number of these by γ. By the suppressed Bernoulli model, these have the distribution (14) (Casella and Berger 1990, p. 96); note that γ includes one less "positive run" than γ^*. After allowing for these positive ranks, the remaining $n-\gamma$ of the ranks are independently positive and negative, leading to the final term in (13).

Recalling that W_m^+ has mean and variance given for fixed m by $E[W_m^+] = m(m+1)/4$ and $\operatorname{var}[W_m^+] = m(m+1)(2m+1)/24$, and using the fact that $E[\gamma] = k_0$ and $E[\gamma^2] = k_0^2 + 2k_0$, we find that

$$\begin{aligned}E[T_n] &= E_\gamma[E[W_{n-\gamma}^+ + (n-\gamma+1) + \cdots + n|\gamma]] \\ &= \sum_{\beta=0}^{\infty}\left[\frac{(n-\beta)(n-\beta+1)}{4} + \frac{n(n+1)}{2} - \frac{(n-\beta)(n-\beta+1)}{2}\right]P[\gamma=\beta] \\ &= \frac{n(n+1)}{4} - \frac{(2n+1)}{4}\,E[\gamma] + \frac{1}{4}\operatorname{var}[\gamma] + \frac{1}{4}\,[E[\gamma]]^2 \end{aligned}$$

$$+ (n+\tfrac{1}{2})E[\gamma] - \tfrac{1}{2}E[\gamma^2]$$

$$= \frac{n(n+1)}{4} + \frac{k_0[2n-k_0-1]}{4}. \quad \text{(A.5)}$$

Note that this shows that

$$E[T_n] = E[W_n^+] + \frac{k_0[2n-k_0-1]}{4}. \quad \text{(A.6)}$$

A similar calculation shows that if we actually know the value of γ, then

$$E[T_n|\gamma] = E[W_n^+] + \frac{\gamma(2n-\gamma+1)}{4}.$$

Using this, for the variance we have

$$\begin{aligned}
\mathrm{var}[T_n] &= E_\gamma[\mathrm{var}[T_n|\gamma]] + \mathrm{var}_\gamma[E[T_n|\gamma]] \\
&= \sum_{\beta=0}^{\infty} \mathrm{var}\left[W_{n-\beta}^+ + (n-\beta+1) + \cdots + n]P[\gamma=\beta]\right. \\
&\qquad \left. + \mathrm{var}_\gamma[E[W_n^+] + \frac{\gamma(2n-\gamma+1)}{4}\right] \\
&= \sum_{\beta=0}^{\infty} \mathrm{var}[W_{n-\beta}^+]P[\gamma=\beta] + \mathrm{var}_\gamma\left[\frac{\gamma(2n-\gamma+1)}{4}\right] \\
&= \frac{1}{24}\left(\sum_{\beta}(n-\beta)(n-\beta+1)\right. \\
&\qquad \left. \times (2(n-\beta)+1)P[\gamma=\beta]\right) \\
&\qquad + E\left[\left[\frac{\gamma[2n-\gamma+1]}{4}\right]^2\right] - \left[E\left[\frac{\gamma[2n-\gamma+1]}{4}\right]\right]^2 \\
&= \frac{1}{24}\,[2n^3+3n^2+n-[6n^2+6n+1]E[\gamma] \\
&\qquad + [6n+3]E[\gamma^2] - 2E[\gamma^3]] \\
&\qquad + \frac{1}{16}\,[E[\gamma^4] - [4n+2]E[\gamma^3] + [4n^2+4n+1]E[\gamma^2]] \\
&\qquad - \left[\frac{1}{4}\,[[2n+1]E[\gamma] - E[\gamma^2]]\right]^2 \\
&= \mathrm{var}[W_n^+] + \frac{1}{24}\,[10k_0^3 + 27k_0^2 + 17k_0 - 18nk_0^2 \\
&\qquad -18nk_0 + 6n^2k_0], \quad \text{(A.7)}
\end{aligned}$$

where we have used the fact that $E[\gamma^3] = k_0^3 + 6k_0^2 + 6k_0$ and $E[\gamma^4] = k_0^4 + 12k_0^3 + 36k_0^2 + 26k_0$. This gives our result.

Remark. Note that in using this approach, we are assuming that n is sufficiently larger than k_0 so that γ will not be truncated prior to actually finding k_0 negative values. In practice, this seems a reasonable assumption given that we should not consider such methods if k_0 were overly large with respect to n. We have also ignored the fact that we know that $\gamma \leq \gamma^*$; using the distribution of γ conditioned on this inequality would be more accurate, although the algebra involved would be considerably more tedious, and the results indicate that the estimators we use have acceptable properties without this extra level of complication.

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